JIACHENG GAO

Zhejiang Province, China | jcgao@smail.nju.edu.cn | Google Scholar | Homepage

RESEARCH INTEREST

Zero-knowledge Proofs, Secure Multiparty Computation, Cryptography, Security

EDUCATION

Nanjing University, Nanjing, China 09/2024 - 06/2027
Major: Master of Science in Computer Science and Technology
Nanjing University, Nanjing, China 09/2020 - 06/2024
Major: Bachelor of Science in Computer Science and Technology | **GPA: 4.50/5.0**
Related Course: Discrete Mathematics (98), Elements of Cryptography (96), Graph Theory and Algorithms (98), Formal Languages and Automata (96), Combinatorics (94), Computational Complexity (97), Advanced Algorithm (91)

RESEARCH EXPERIENCE

Intern, Research Group of Jiaheng Zhang, NUS. Jul 2025 - Feb 2026

- Designed and implemented a zero-knowledge proof framework for verifiable CNN training based on coding-based multilinear PCS (Orion) and the sum-check protocol.
- Proposed a new batch-commitment scheme for multilinear polynomials, reducing proof size and amortizing commitment costs.
- Developed a ZKP protocol for training and inference of gradient-boosting decision trees (GBDT) using univariate PCS (KZG) and lookup arguments.
- Proposed new batching tools for univariate polynomial commitments, and new approaches to commit GBDT with constant training/inference proof size.
- Co-authored two papers; one under review, the other accepted to ACM CCS 26.

M.S. Researcher, Research Group of Yuan Zhang, NJU. Sep 2024 - present

- Proposed a generic transformation from permutation protocols to shuffle protocols with linear online communication and computation complexity, whereas prior work incurs quadratic complexity.
- Optimized the shuffle protocol under Shamir secret sharing and implemented a prototype system with empirical evaluation.
- Designed MPC protocols for small-element operations, including one-hot encoding, comparison, digit decomposition, and sorting.
- Co-authored three papers (details in publication section).

PUBLICATIONS

Zero-Knowledge Proofs for Gradient Boosted Decision Trees
***Jiacheng Gao**, Wenjie Qu, Yuan Zhang, Sheng Zhong, Jiaheng Zhang*
ACM CCS 26; ePrint Link

Efficient SNARK for Convolutional Neural Network Training
Wenjie Qu*, **Jiacheng Gao***, Jiaheng Zhang
Manuscript in preparation

UFOs: An Ultra-fast Toolkit for Multiparty Computation of Small Elements
Jiacheng Gao, Moyang Xie, Yuan Zhang, Sheng Zhong
Manuscript in preparation; ePrint Link

SLIDE: Shuffle Shamir Secret Shares Uniformly with Linear Online Communication and Guaranteed Output Delivery
Jiacheng Gao, Moyang Xie, Yuan Zhang, Sheng Zhong
Journal of Computer Security (JCS); ePrint Link

Free Linear Online Phase for Secure Multiparty Shuffle
Jiacheng Gao, Yuan Zhang, Sheng Zhong, Changyu Dong
ASIACRYPT 2026; ePrint Link

Revisiting Privacy-Preserving Min and k-th Min Protocols for Mobile Sensing
Jiacheng Gao, Yuan Zhang, Sheng Zhong
IEEE Transactions on Dependable and Secure Computing (TDSC), 2023. Link

HONORS & AWARDS

Outstanding Graduate of Nanjing University	2024
Elite Program Scholarship, Grand Award	2023
Elite Program Scholarship, Excellence Award	2021 and 2022
People Scholarship, Third Prize	2021 and 2022
National Olympiad in Informatics (Zhejiang Province), First Prize (Top 1%)	2019

SKILLS

Programming: C++, Python

Math: Abstract Algebra, Information Theory, Complexity Theory